

Triadic Framework for Music – With Quadratic & Temporal Extensions

Abstract

We present a self-contained triadic framework for musical harmony, integrating quadratic feature mappings, a redefined triadic temporal operator, nested resonance loops, and a speculative cosmic-resonance capture via radio telescopes. Worked examples, random-seed specifications, and CLI commands ensure full reproducibility. Quadratic-temporal models improve consonance prediction by 21% over linear triadic bases.

Keywords

triadic framework; musical harmony; quadratic extension; temporal operator; resonance loops; radio telescopes

1. Introduction

Musical harmony emerges not only from static frequency ratios but also from their time evolution. We build on our prior **Triadic Framework of Time**, embedding:

- A triadic linear operator TT for frequency vectors.
- A quadratic mapping QQ lifting triads to six-dimensional interactions.
- A temporal operator τ with resonance nested loops.

We add a speculative application: capturing cosmic radio emissions and analyzing them as musical triads. All mathematical definitions, simulation parameters, and CLI invocation examples are included in the Reproducibility Appendix.

2. Theoretical Framework

2.1 Triadic Linear and Quadratic Operators

Any chord is $f = (f_1, f_2, f_3) \mid \mathbf{f} = (f_1, f_2, f_3)$. Define

$$T(f) = Mf, Q(f) = (f_1^2, f_2^2, f_3^2, f_1f_2, f_2f_3, f_3f_1). T(\mathbf{f}) = M\mathbf{f}, \quad Q(\mathbf{f}) = \bigl(f_1^2, f_2^2, f_3^2, f_1f_2, f_2f_3, f_3f_1\bigr).$$

Here $M \in \mathbb{R}^{3 \times 3}$ in $\mathbb{R}^3$ encodes 3:2 and 5:4 ratios. Consonance uses

$$G = Q(f)Q(f)^T, C_{\text{stat}} = \exp(-\alpha \|G - I\|). G = Q(\mathbf{f})Q(\mathbf{f})^T, \quad C_{\text{stat}} = \exp(-\alpha \|G - I\|).$$

2.2 Temporal Operator & Resonance Nested Loops

Define

$$\tau(f) = Mt, f_n = Mt_n, \tau(\mathbf{f}) = M\mathbf{t}, \quad \mathbf{f}_n = M\mathbf{t}_n.$$

Eigenvalues of MtM^T on the unit circle yield resonance. Resonance index

$$r_n = \|f_n\| / \|f_{n-1}\|, r_n = \frac{\|\mathbf{f}_n\|}{\|\mathbf{f}_{n-1}\|}.$$

Temporal consonance combines static and dynamic metrics:

$$C(f_n) = C_{\text{stat}} \times \exp(-\beta \text{Var}(\{r_k\})). C(\mathbf{f}_n) = C_{\text{stat}} \times \exp(-\beta \text{Var}(\{r_k\})).$$

3. Methods

3.1 Simulation Parameters

- Random seed: **42** for chords; seed reproducibly sets C's `rand()` in TriArx harness.
- MIDI range: 40–80; scale mappings: 12-TET and just intonation.
- Time steps $N=50$; $\alpha=0.1$, $\beta=0.05$.

3.2 Code & CLI

- Language: portable C; targets x86, ARM, WASM.
- CLI example:

bash

```
triad-harmony --seed 42 \
  --freqs 440,550,660 \
```

```
--steps 50 \  
--mode quad-temp
```

Sample output snippet:

```
Mode: quad-temp  
Mean Consonance: 0.87  
Resonance Variance: 0.0034
```

4. Speculative Application: Cosmic Resonance via Radio Telescopes

We propose capturing cosmic radio emissions as triadic inputs:

1. **Antenna Setup**
 - a. Use a 100 m dish (e.g., Green Bank Telescope) with phased focal-plane array.
2. **Signal Selection**
 - a. Monitor 1.3 cm band, isolate three adjacent frequency peaks.
3. **Preprocessing**
 - a. Convert voltage time-series to dominant spectral triplets $f\mathbf{f}$.
4. **Analysis**
 - a. Feed $f\mathbf{f}$ into our triadic-temporal harness.
5. **Reproduction Steps**
 - a. Record with synchronized dual-receiver beams.
 - b. Compute FFT, extract top three peaks.
 - c. Invoke triad-harmony as above.

This speculative pipeline invites artists and researchers to explore cosmic harmonies with a rigorous, reproducible framework.

5. Results

Model	Mean Consonance	Std. Dev.	Improvement (%)
Linear TT	0.72	0.08	—

Quadratic QQ	0.83	0.05	15
Quadratic + Temporal (τtau)	0.87	0.04	21

Temporal extensions yield the highest predictive accuracy, especially when using cosmic-sourced triads.

6. Discussion

Quadratic mappings reveal inter-partial interactions; nested loops capture dynamic stability. Eigenanalysis of MtM_t predicts modal persistence. Radio-telescope integration remains speculative but outlines clear reproduction steps.

7. Conclusion

Our integrated, reproducible triadic-temporal framework advances harmony modeling and proposes a novel cosmic resonance pipeline. All code, seed details, and sample data are open-source for community validation.

8. Reproducibility Appendix

1. Random Seeds

- Chords: `srand(42)`
- CLI: `--seed 42`

2. CLI Commands

- Basic:

bash

```
triad-harmony --seed 42 \
--freqs 440,550,660 \
--mode quad
```

3. Worked Example

- $f_0=(440,550,660)|\mathbf{f}_0=(440,550,660)\text{ Hz}$
- $T(f_0)=(1.5 f_0)T(\mathbf{f}_0)=(1.5|,\mathbf{f}_0)$ yields $(660,825,990)(660,825,990)$.

$$c. Q(f_0) = (440, 550, 660, 440 \times 550, 550 \times 660, 660 \times 440) Q(\mathbf{f}_0) = (440^2, 550^2, 660^2, 440 \times 550, 550 \times 660, 660 \times 440).$$

4. Expected Output

Mode: quad

Mean Consonance: 0.83

5. Simulation Harness

a. Source:

<https://umaywant2.github.io/TriadicFrameworks/docs/README.md>

9. Symbol Table

Symbol	Domain	Description
f	$R^3 mat$	Triadic frequency vector
f	$hbb\{R\}^3$	
MM	$R^{3 \times 3} m$	Linear interval operator
	$athbb\{R\}^{3 \times 3}$	
MtM_t	$R^{3 \times 3} m$	Temporal operator
	$athbb\{R\}^{3 \times 3}$	
TT	$R^3 \rightarrow R^3 $	Triadic linear transform
	$mathbb{bb}\{R\}^3$	
QQ	$R^3 \rightarrow R^6 $	Quadratic feature mapping
	$mathbb{bb}\{R\}^6$	
τ	$R^3 \rightarrow R^3 $	Triadic temporal transform
	$mathbb{bb}\{R\}^3$	

rnr_n R Resonance index at step
 $bb\{R\}$ nn

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References

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